

Layer 1 Pipeline — Full Command List

Run in this order. Paths shown match your actual folder structure (Result_data, Result_data_organized, Trained_data, Mapping, Classifying_phase/...).

1. Reorganize labeled data into Phase/Top/Earth_wire/Bottom

```
cd Training_phase/Preparation_data
python reorganize_tower_folders.py --root ../../Result_data --output ../../Result_data_organized --dry_run
python reorganize_tower_folders.py --root ../../Result_data --output ../../Result_data_organized
```

2. Pool per-tower classes into one flat training folder

```
cd Mapping
python build_training_dataset.py --root ../Result_data_organized --output ../Trained_data --dry_run
python build_training_dataset.py --root ../Result_data_organized --output ../Trained_data
```

3. Extract features (with augmentation) and build the (X, y) training pair

```
cd Mapping
python build_xy.py --data_root ../Trained_data --augment_per_image 4 --output_prefix build
```

4. Train the classifier

```
cd Mapping
python train_classifier.py
```

Uses build_X.npy / build_y.npy from step 3. Saves layer1_model.joblib.

5. Classify new, unlabeled tower photos

```
cd Mapping
python classify.py --root ../Classifying_phase/Input_data --model layer1_model.joblib --output
../Classifying_phase/Classified_output --dry_run
python classify.py --root ../Classifying_phase/Input_data --model layer1_model.joblib --output
../Classifying_phase/Classified_output
```

6. Reorganize predictions back into per-tower structure

```
cd Mapping
python sort.py --input ../Classifying_phase/Classified_output --output
../Classifying_phase/Sorted_classified_output_by_tower --dry_run
python sort.py --input ../Classifying_phase/Classified_output --output
../Classifying_phase/Sorted_classified_output_by_tower
```

Note before running Step 4

train_classifier.py currently loads step4_X.npy / step4_y.npy from an earlier version of the pipeline, but build_xy.py now saves as build_X.npy / build_y.npy. Update the load lines in train_classifier.py to match before running Step 4, or the file names will not line up.